

MMIE/MMME– 201 Product Design and Life cycle Management

Unit1 Product Design: Product specifications, concept development, configuration design involving synthesis, analysis and optimization, Detailed design, Presentation of design, Oral and Visual presentations, various types of models used in product design, Design through creative routes, Adaptive and variant design, Concurrent Engineering theory.

Unit2 Product Life cycle Management definitions, Product data management, Evolution of PLM, Pre-requisites for PLM, PLM benefits, PLM implementation methods, Product Life cycle Engineering: Design for X, Components of DFX.

Unit 3 Design for manufacturing and Design for assembly, Role of Aesthetics and Ergonomics in design, Robust Design using Taguchi methods, Reliability based design, Modular versus integral design, Design for environment (DFE), Life cycle assessment (LCA), Steps of LCA, Determining system boundaries, Life Cycle Inventory compilation, Common LCA methods and their impact categories, Applying DFE through LCA, Case studies.

Unit 4 Value analysis: scope, techniques and job plan, Standardization, Renard series, Simplification versus Variety in products, Patents, copyright and Intellectual Property Rights.

Unit 5 Marketing Management: Philosophies of Marketing, Market and Product strategies, BCG matrix, Portfolio management, New Product development strategy. Marketing channels, Pricing strategies and Promotional strategies, Consumer behavior, Sales Management, Planning of sales, Sales skills, evaluation and promotion, Advertising methods, preparation of advertising briefs.

References:

1. John Stark; Product Life Cycle Management: Paradigm for 21st Century; Springer-Verlag.
2. Grieves Michael, Product Life Cycle Management, McGraw Hill.
3. Chitale A.K. and Gupta R.C.; Product Design and Manufacturing, PHI.
4. Gupta V., Lal G.K. and Reddy; Fundamentals of Design and manufacturing; Narosa Publishing.
5. James Garrat; Design & Technology; Cambridge University Press
6. Dieter; Engineering Design Marketing Management; PHI.
7. Kolter, Philip; Marketing Management; PHI.
8. Stanton; Principles of Marketing, Prentice Hall.
9. Product Life Cycle Engineering and Management, CEP Lecture notes, Prof B Ravi, IIT Bombay

MMIE/MMIP/MMCM/MMME–202 Total Quality Management

UNIT- I

Total Quality Management, Evolution of TQM, Historical perspective, People involvement, Teamwork, Discipline, Supplier involvement, Defining the immediate customer, Quality at source. Various quality awards, Total employee involvement, Competitive Benchmarking.

UNIT-II

Elements of TQM : Learning from Quality Gurus: Edward Deming, Joseph M. Juran, Ishikawa, Philip B. Crosby, etc. and their Applications in Today's Business Environment, Ishikawa Diagram, FMEA, FMECA, Just-in-time philosophy, 5S, TPM, Kaizen and continuous improvements, Life cycle costing.

UNIT-III

Place of quality control in industries, Quality control organization, inspection and quality control. Cost and Economics of Quality systems. Statistical Process Control, Design and application of control charts for variable and attribute. Acceptance Sampling: Various Sampling plans. Selection of sampling plans for different situations. Applications of quality control in industries.

UNIT-IV

Quality Assurance, Process capability studies. Fundamental concepts of Six sigma, General methodology of Six Sigma. Latest developments in six sigma methodology, six-sigma and cost control, ISO-9000; ISO- 14000: Concepts, Certifications, Methods and Implementations.

UNIT- V

Application of TQM to service type organizations, Service guarantees, Case studies on application of TQM to service type organization, QFD, Reliability: Distributions encountered in controlling Reliability, MTTR, MTTF, Failure density, Measurement and Tests, Maintenance and Reliability, Life testing. Pitfalls and difficulties in implementation TQM and Methods of Avoiding Them

Books & References Recommended:

1. TQM by Dale H Bester field Pearson Quality Control by Bester field Pearson Dhillon, Reliability, Wiley Eastern
2. Grant E.L. and Leave Worth, Statistical Quality Control, TMH.
3. Amitava Mitra; Fundamentals of Quality Control and Improvement; Pearson; 2001 K Sridhar Bhatt, TQM, Himalaya Publications. 2005
4. J.M. Juran and Frank M.Gryna (Edts.), Juran's Quality Control Handbook, New York, McGraw Hill Publications, 2001.
5. Tapan P. Bagchi, ISO 9000: Concepts, Methods and Implementation, Allahabad, Wheeler Publishing, 1996.

MMME–203 Logistics & Supply Chain Management

Unit 1 Introduction to SCM: Understanding supply chain, supply chain drivers and obstacles, inbound and outbound supply chain network, Performance Measures and supply chain performance, Decisions in the Supply Chain, Models for Supply Chain Decision-Making.

Unit 2 Supply Chain Inventory Management, demand forecasting in supply chain, aggregate planning in supply chain; planning supply and demand; managing predictable variability in supply, transport and demand, Economic Order Quantity Models, Reorder Point Models, Multi-echelon Inventory Systems, Inventory Optimization, JIT in SCM.

Unit 3 Mathematical Foundations of Supply Chain Solutions; Use of Stochastic Models and Combinatorial Optimization in Supply Chain, Planning, Supply Chain Facilities Layout, Capacity Planning, Dynamic Routing and Scheduling, Managing economies of supply chain; managing uncertainties in a supply chain; determining optimal levels of product availability.

Unit 4 Logistics Management: Definition, Logistics role in the economy and in the firm, Components and requirements, Organization of Logistics functions, Integrating Logistics functions in overall organization structure, Measurement of performance of Logistics function and functionaries, Supply Management and Logistics - Integrated Logistics Planning; Evolution of World Class Management and implication for supply Management, Business Logistics Concept; Customers Satisfaction, Customers Value Creation, Customer Relationship Management, Cost and relationship, Lean logistics.

Unit 5 Transportation, Network Design and Information Technology: Transportation Fundamentals Transportation Decisions, facility Decision; Network design in a supply chain; Information technology and its use in supply chain, Coordination and E-business in a supply chain; financial evaluation in a supply chain, Relation to ERP, E-procurement, E-Logistics, Internet Auctions, E markets, Electronic business process optimization

References:

- 1 Hopp WJ, Spearman, Irwin; Factory Physics: Foundations of Manufacturing; TMH
- 2 Viswanadham Nand Kluwer; Analysis of Manufacturing Enterprises; Academic Pub
- 3 Sridhar Tayur, Ganeshan; Quantitative Models for SCM; Academic Publishers
- 4 Walker William T; Supply Chain Architecture; CRC press
- 4 Handfield RB and Nichols; Introduction to SCM; PHI
- 5 Chopra Sunil and Meindel P; SCM: Strategy, Planning, and Operation; PHI
- 6 Shapiro JF, Duxbury; Modeling the Supply Chain; Thomson Learning,
- 7 Levi DS, Kaminsky, Levi; Designing and Managing the Supply Chain...;TMH
- 8 Chitale AK and Gupta RC; Materials Management; PHI
- 9 Ronald H. Bolau "Business Logistics Management",
- 10 Bowersox D.J., Closs D.J. & Helferich O.K "Logistical Management",
- 11 Lambert D.M. & Stock J.R "Strategic Logistics Management",

MMIE/MMME–204 Industrial Organization and Management

Unit 1 Introduction: Definitions of management, administration and organization, management functions- planning, organizing, leading and controlling, organization performance, management skills, Mintzberg's three role-category of informational, interpersonal/ HRM and decisional, ten roles of manager, managing in crisis, historical evolution of management thinking: classical, humanistic, modern perspectives and jungle of theories, scientific, bureaucratic, administrative, human resources, behavior, system and contingency theories, learning organization, corporate/ organization cultures, symbols, stories, heroes, slogans and ceremonies, Informal organizations, ethics and social responsibilities.

Unit 2 Planning: Definition, why planning is necessary and difficult, mission, vision, goals, objectives, strategies, plans and schedules, three levels hierarchy of planning and goals, contingency plans, management by objectives (MBO), BCG matrix, SWOT analysis, Porters five factor competency, **Organization:**-Definition, dimensions of organizations-size/specialization, behavior formalization and authority centralization, departmentalization, span and line of control, responsibility, authority (to use resources) and accountability, line, staff, matrix and virtual network organization, Minzberg's organization typology with techno-structure and support staff, horizontal coordination by task force and project teams, informal groups, Business Process Reengineering, effect of environment (stable, unstable), strategy (stable/cost-leadership/efficiency, flexible/differentiation/innovative) and Technology (job-shop, batch, mass, lean, TQM, e-business) on structure.

Unit 3 Leading and decision making: types of decision-structured (programmed) and unstructured, decision under risk, uncertainty and ambiguity, models like rational/normative/classical, bounded-rationality/satisficing/administrative and political, six rational decision making steps, personal differences/**styles**- directive, analytical, conceptual and behavioral, Vroom-Jago's five leader/ group participation styles of decide, consult-individual/ group, facilitate, delegate and the model to decide the styles, new speedy decision processes of e brainstorming, learning by failures at early, debate and five W's, management in crisis/ turbulence and change management.

Unit 4 Controlling: mutual relation in plan and control, control loop, administrative and financial controls, budgets, types of Ownership, Individual, Partnership: - Joint stock company, Public Ltd and Pvt. Ltd companies, cooperative organization, cooperative Societies, public sector organization, comparative evaluation of different forms of Business ownership.

Unit 5: System Concepts: Types, definition & characteristics; supra & subsystems, key component; boundary & interface complexity; feedback (pull) & feed forward (push) controls, open flexible-adaptive system, computer as closed system, law of requisite variety; system coupling, stresses and entropy; functional & cross functional system; Steven Alter's nine element work system model and its comparison with IPO (input- processing-output) model, structure and performance of work systems leading to customer delight.

References:

- 1- DaftR; The new era of management; Cengage.
- 2- Bhat Anil, Aryakumar; Management: Principles, ProcessesPractices; Oxford higher edu.
- 3- Robbins SP; Organization theory; PHI
- 4- Robbins SP, Judge TA; Organizational Behavior; PHI
- 5- Agrawal RD; Principles of Organization.
- 6- Davis & Olson; Management Information System; TMH.
- 7- Steven Alter; Information systems,Pearson,

MMIE/MMME–205 Management Information System, ERP & CRM

UNIT 1 Management Information System (MIS) definition, Objectives and benefits, MIS as strategic tool, obstacles and challenges for MIS, functional and cross functional systems, hierarchical view of CBIS, structured and unstructured decision, Operation and mgt support, Decision process and MIS, info system components and activities, Value chain and MIS support.

UNIT 2 System concepts: types, definition, characteristics, feedback (Pull) and feed-forward (Push) control, system stress and entropy, computer as closed system, law of requisite variety, open and flexible (Adaptive) systems, work system model and comparison with input-process-output model, five views of work system: structure, performance, infrastructure, context and risk and their effect on product performance.

UNIT 3 Info concepts: define data, info, knowledge, intelligence and wisdom. Information characteristics and attributes, info measurement and probability, characteristics of human as info processor.

UNIT 4 Planning and control Concepts: terminologies, difficulties in planning, system analysis and development plan-purpose and participants, info planning, (SDLC) system development life cycle for in-house and licensed sw, system investigation, analysis of needs, design and implementation phases, training of Operational personnel, evaluation, Control and Maintenance of Information Systems.

UNIT 5 E-business components and interrelationship, Evolution of Enterprise Resource Planning (ERP) from MRP, Supply chain management (SCM) and Customer relationship management (CRM), Integrated data model, strategic and operational issues in ERP, Business Process Re- Engineering (BPR), significance and functions of BPR, information technology and computer NW support to MIS.

UNIT 6 ERP Implementation, role of consultants, vendors and users, customization, methodology of ERP implementation and guidelines for ERP implementation, ERP modules.

References:

- 1.Davis and Olson, MIS, TMH
- 2.James O'Brian, MIS,TMH
- 3.Oz, MIS, Cengage
- 4.Business Process Re-Engineering, Jayaraman, TMH.
- 5.ERP by V.K.Garg, PHI
- 6.ERP by Alex Leon.
- 7.ERP, Monk, Cengage (Thomson)
- 8.Manuals of SAPP, MFG-pro